

LEVERAGING PREDICTABILITY OF DEEP LEARNING TECHNIQUES ON FINANCIAL TIME SERIES FORECASTING

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ABSTRACT

Financial time series forecasting remains a challenging task due to the inherent volatility and unpredictability of markets, particularly in emerging economies such as India. Traditional statistical models and machine learning approaches struggle to capture the complex, non-linear dependencies in financial data. In this study, we propose a model integrating Bidirectional Long Short-Term Memory (Bi-LSTM) and Dense Networks, to enhance predictive accuracy. The Bi-LSTM component effectively captures sequential dependencies and long-term trends, while the Dense branch extracts meaningful non-temporal relationships, leading to a richer representation of market patterns.

Extensive experimentation with baseline, hybrid deep learning models, and probabilistic approaches demonstrated the challenges posed by fluctuating stock market dynamics. To address model interpretability, we incorporate Explainable AI (XAI) techniques, ensuring transparency in feature significance and model decisions. Our results highlight the advantages of combining sequential and non-sequential learning paradigms for financial forecasting, providing deeper insights into stock market trends. This research contributes to the growing field of AI-driven financial modeling by identifying an effective approach for handling high-variance stock market data.

INTRODUCTION

Time series forecasting predicts future values based on past trends, crucial for financial markets. Stock prices are highly volatile and influenced by multiple factors, making prediction challenging. Traditional models capture trends but struggle with sudden fluctuations. This project integrates Bi-LSTM for sequential dependencies and a Dense Network for non-sequential patterns. Explainable AI (XAI) enhances transparency, making predictions more interpretable for analysts.

Time series data consists of observations recorded at regular time intervals, exhibiting patterns such as trends, seasonality, and cyclic variations. Financial time series, like stock prices and exchange rates, are particularly challenging due to their non-linearity and high volatility.

Forecasting involves predicting future values based on past observations, mathematically expressed as:

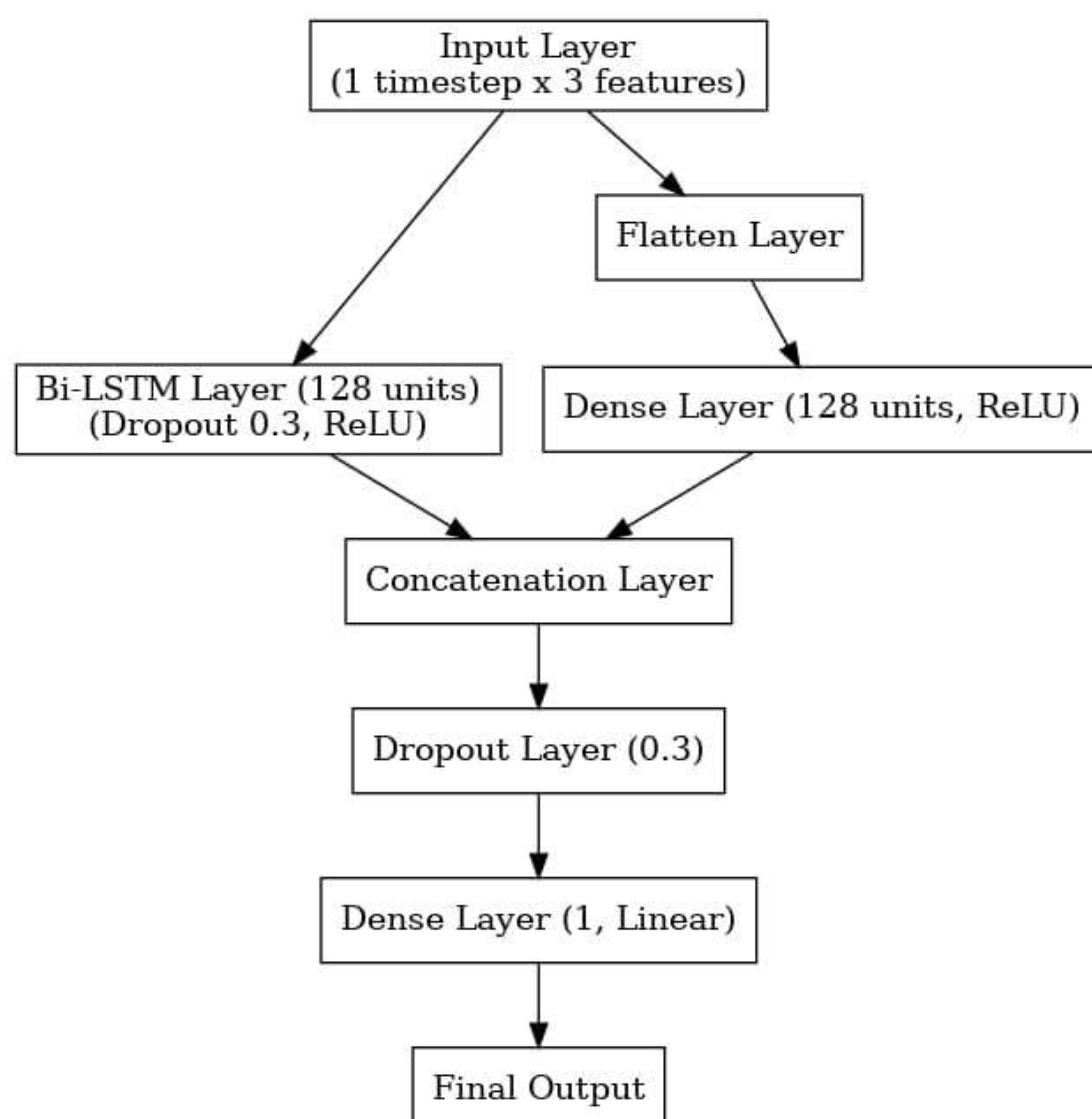
$$Y_t = f(t) + \epsilon_t$$

where Y_t is the observed value, $f(t)$ represents the underlying trend, and ϵ_t is random noise. Traditional methods like Autoregressive (AR) and Moving Average (MA) capture past dependencies but struggle with abrupt market shifts. This study leverages a hybrid deep learning approach, integrating Bi-LSTM for sequential dependencies and Dense Networks for non-sequential patterns, improving forecasting accuracy in dynamic financial environments.

The project focuses on forecasting closing price in stock within the Indian financial market, using historical data from the BSE SENSEX NEXT50 (2014–2024). This dataset includes major market fluctuations, allowing the model to learn from diverse financial conditions. The study evaluates deep learning techniques, emphasizing the hybrid model's ability to enhance prediction accuracy and interpretability through XAI.

The model enhances stock prediction, portfolio management, and algorithmic trading, optimizing asset allocation and automating decisions. It supports risk assessment and aids economic research, helping policymakers monitor financial stability.

METHODOLOGY



ALGORITHM

The proposed model enhances stock price prediction by integrating Bidirectional Long Short-Term Memory (Bi-LSTM) networks with dense feature extraction layers. This combination allows the model to effectively capture both temporal dependencies and feature-based representations, resulting in improved forecasting accuracy.

The methodology begins with data preprocessing, where historical stock prices are normalized and structured into time-series sequences. Missing values are imputed to ensure data continuity.

The model architecture features a Bidirectional LSTM layer with 128 units to learn dependencies in both forward and backward directions, enhanced by ReLU activation and dropout for regularization. A parallel dense path extracts features from the flattened input sequence. These outputs are concatenated and passed through additional dropout layers to prevent overfitting. The final dense layer with a linear activation function generates the stock price predictions.

Hyperparameters such as the number of LSTM units, dropout rate, and learning rate are optimized through Bayesian Optimization, which systematically explores the hyperparameter space to find the optimal settings. The model is trained using the Adam optimizer and Mean Squared Error loss function, with dynamic learning rate adjustments and early stopping to ensure robust performance. Evaluation metrics like MAE, MSE, RMSE, and MAPE are used to assess model accuracy on a separate test dataset, ensuring reliable predictions for real-time trading scenarios.

RESULTS

METRIC	PROPOSED MODEL	BEST VALUE FROM BASELINE MODELS	BEST MODEL FROM BASELINE MODELS
MAE	399.0173	397.4080	GRU
MSE	301999.742	306645.694	GRU
RMSE	549.545	553.7559	GRU
MAPE	0.469	0.05	CNN 2D

The Actual vs. Predicted Stock Closing Prices graph shows the model's predictions closely following actual market trends, indicating strong generalization. The Error Distribution Curve reveals a bell-shaped error distribution centered around zero, suggesting random errors with no directional bias. The SHAP Summary and Dependence Plots highlight the significant impact of Open, High, and Low prices on predictions, with Open being the most influential. The LIME Explanation Chart confirms that higher Open, High, and Low prices positively influence the model's predictions.

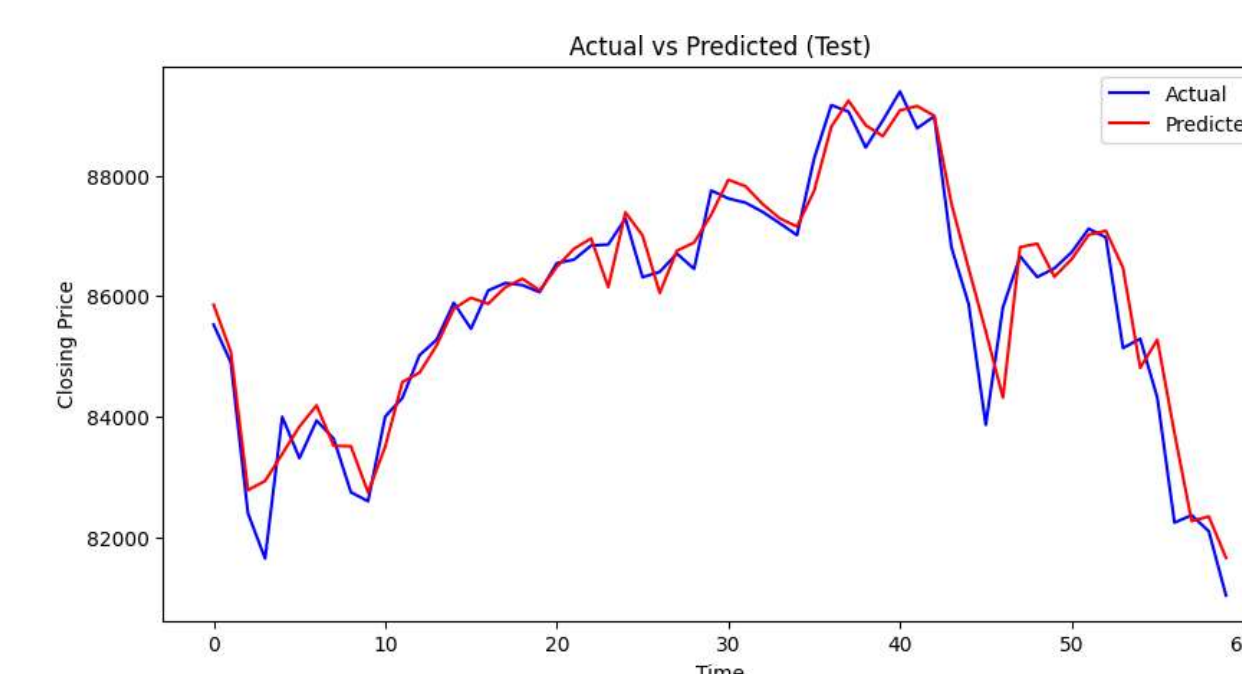


Fig 1: Actual vs Predicted Closing Price values

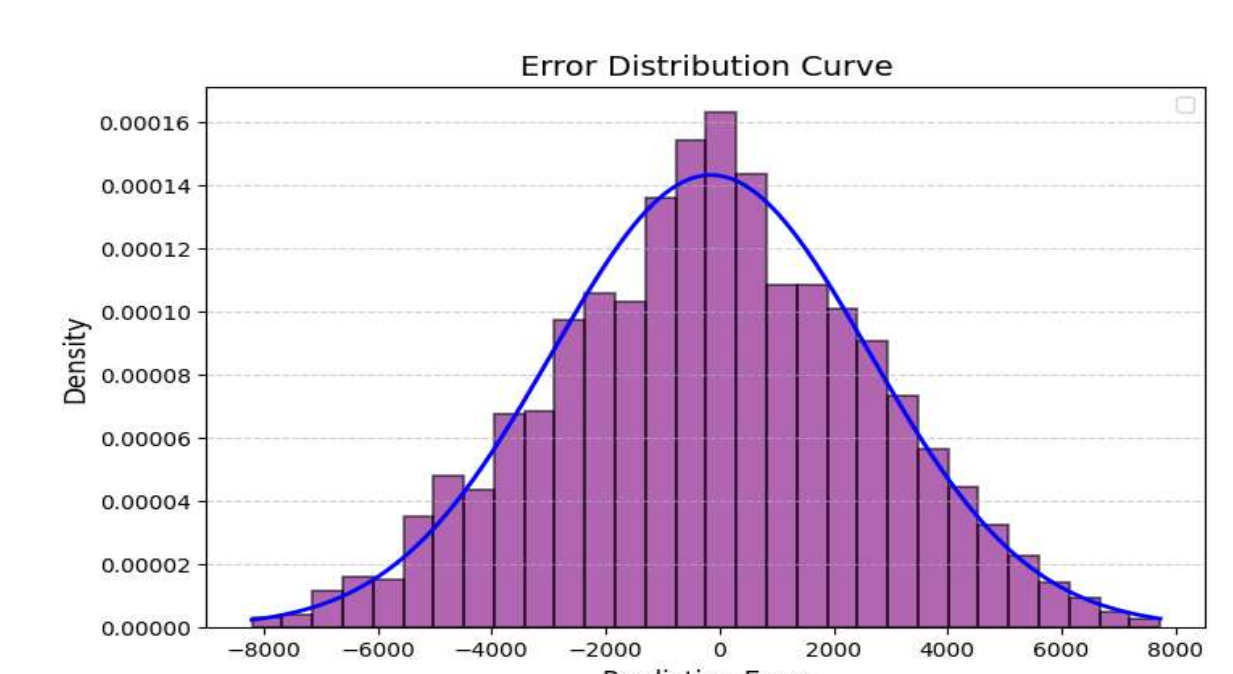


Fig 2: Error Distribution Graph for model's prediction

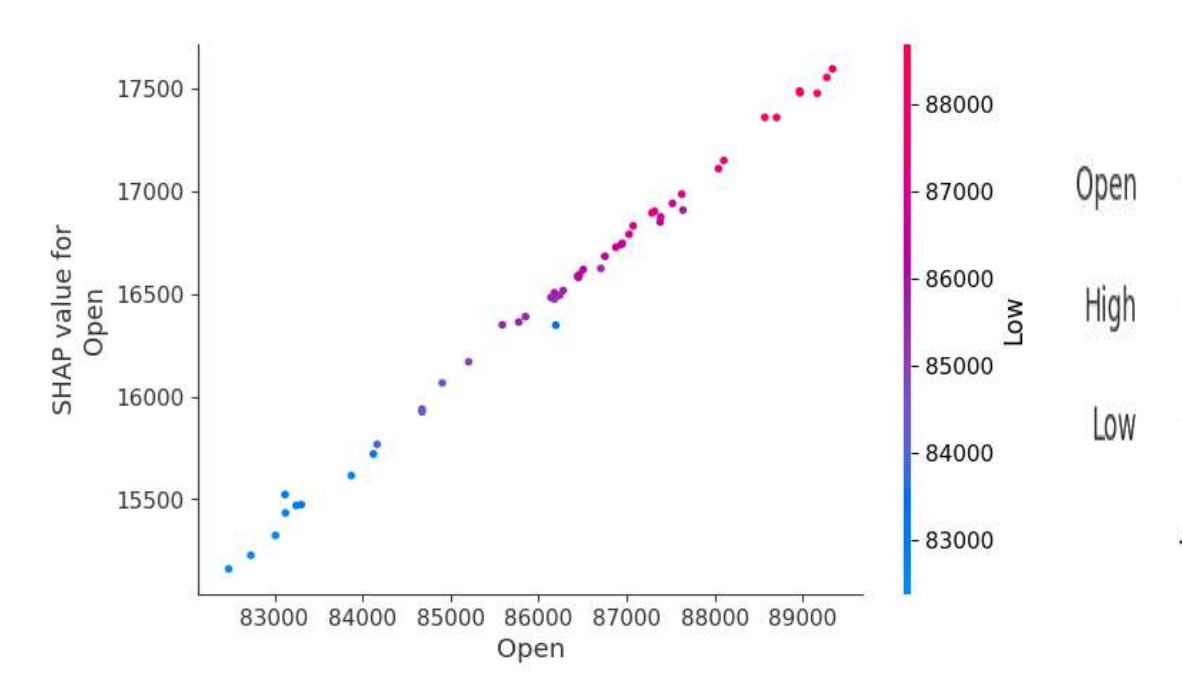


Fig 3: SHAP Dependence Plot

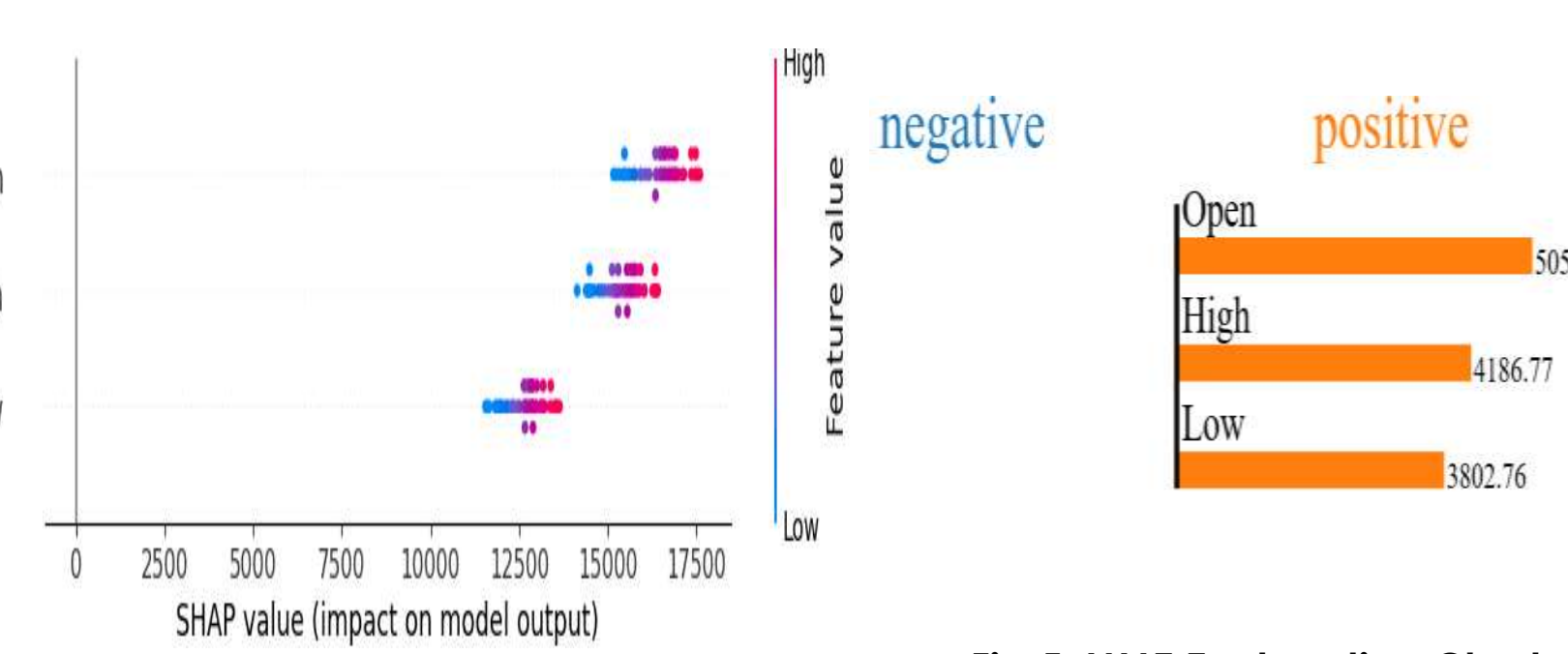


Fig 4: SHAP Summary plot

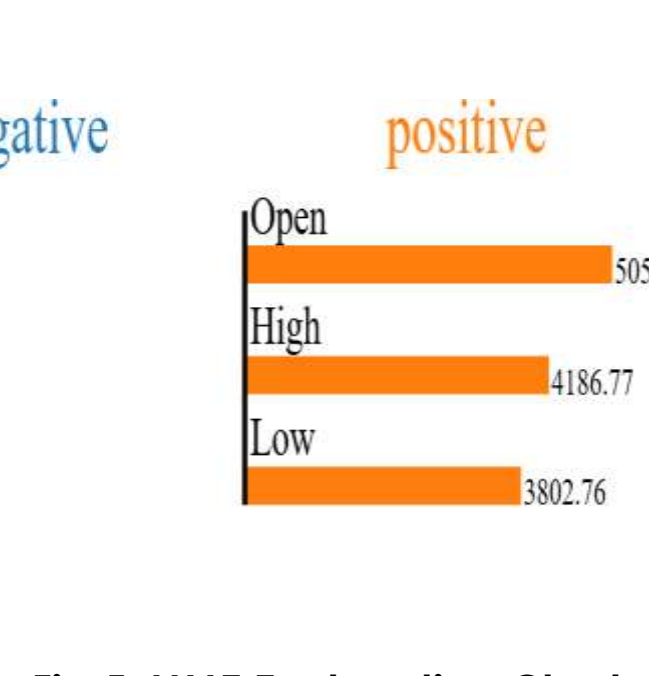


Fig 5: LIME Explanation Chart

CONCLUSION

The proposed Bi-LSTM + Dense Network model demonstrates strong predictive accuracy for stock price forecasting, effectively minimizing errors and avoiding overfitting. It achieves competitive results, with improvements of approximately 1.5% in MSE and RMSE compared to baseline models. The proposed model provides a robust framework for stock price forecasting, balancing accuracy and generalization, and showing potential for further improvements in future work.

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